

Use of Augmented Reality for Computational Thinking Stimulation through Virtual

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Abstract— Studies with augmented reality show that while the technology helps the student learn, it may end up disturbing other points, showing that more research is needed with the use of augmented reality. Also, a topic that is growing within education is Computational Thinking, which says that the thinking like computer scientists is necessary for the 21st century and should be disseminated mainly in schools during basic education. In the face of this, this work created an augmented reality android application that tracks physical tiles and creates 3D maps for the user, also giving a character that can be programmed to walk on this created map, simulating a programming toy. Finally, tests carried out with the application have shown that augmented reality currently has performance problems when multiple targets are used, so, devices with low processing have problems at map tracking. These issues affected the tests with the children, which made it difficult for them to track the map like they wanted, but they still liked the app even after it. That shows that the acceptance of the technology by the children is good, which can help developing Computational Thinking, but the technology limitation is still a problem. These obstacles while currently hindering the use of augmented reality in education may in the future be irrelevant due to the constant advancement of technology. This paper's main contribution is showing one possibility of using AR with Computational Thinking and what needs to evolve for it to be used by the usable by the children.

Keywords— *Computational Thinking, Augmented Reality, Technology in Education*

I. INTRODUCTION

Research on the teaching of scientific and curricular content using augmented reality (AR) has shown positive results in student education, mostly in high schools. [1] [2] [3] [4] [5]. AR deals with a kind of technology that uses reality as a basis for virtual objects to be shown through it with the help of real-time interaction and three-dimensional registration. The studies with RA in education, mainly done by Radu [5], showed that the technology has a lot of educational potential, especially on the motivation of the students as the tasks performed, thus causing a greater aggregation of the knowledge by the students. Part of its negatives may be more related to the application of the activities than the potential of the technology itself.

However, even though AR has been positive in transmitting scientific knowledge, direct teaching of curriculum content is not the only way to educate students. Papert [6] shows the term Computational Thinking which for

him, at the time, was to bring a computer's logic to schools and other learning environments. The computer would give the learner more power because he could use the tools available on the computer to facilitate learning the content.

Computational Thinking brings the idea of the thinking way used in computer science. Some of its concepts are separated into: mental processes (e.g. algorithm design, decomposition, pattern recognition, etc.) and tangible results (e.g. automation, data, generalization of standards, etc.) [6]. Wing [7] describes the various concepts and approaches of Computational Thinking. She describes Computational Thinking as the way the computer scientist thinks. Thinking about the planning, construction, research, interpretation, abstraction, decomposition, automation, simulation and problem solving, as computer scientists do while working on their projects. Raabe [8], presented the RoPE, a programming toy to start children from 4 to 8 years old with math concepts and algorithms. It works from instructions given by the commands on its surface, with the possibility of using pedagogical mats / maps as a place to support the development of stories and challenges.

In this context, this study aims to evaluate if the use of AR to stimulate Computational Thinking can present significant results when used in the creation of virtual maps in activities with programming toys. An application for the Android operating system that uses Augmented Reality was created to interact with markers presented on physical cards and to generate 3D maps with AR according to the position of those cards. Then giving the user a character to program in that map. It is intended to simulate the same interaction that the programming toys, like they generate when playing with it, but with rendering in the RA, offering freedom for the creation of the map.

II. DEVELOPMENT

This work was developed an Augmented Reality application for the Android operating system to verify how Computational Thinking can be stimulated by the technology. The application tracks physical cards that together form a map, this map will be rendered in the Augmented Reality system, and as soon as it finishes, it will enable a user-programmable character to walk this map. This work is divided into 2 parts that were developed in parallel: creation of the physical cards and creation of the application in AR.

A. Physical Cards

Physical cards are parts of a map that a character in RA will walk through. Each card is 1 tile and these cards have different types of designs that can be rendered. The tiles are in the following sequence from left to right and from top to bottom: grass, concrete, rocks, sand, tiger, house, tower, dragon, tree, car, fountain, chargirl and charboy. They can be printed on any material if the image remains complete.

Besides these tiles shown in Fig. 1, there is also the tile base that can be seen in Fig. 2. It is used to render the map after being tracked, that is, it will be where the character programming and the graphic interface will be used, therefore for a better experience, it will have a size of 19.5x19.5cm in this work. Each of the tiles in this project was printed on adhesive paper and glued onto 6x6cm mdf boards. There were 49 boards in mdf, on both sides a tile image was used, so there are 98 printed tiles. Only one of the cards contains the characters, each one being on one side of the board, thus preventing maps with more than one being made.

B. Augmented Reality Application

The application was developed using the Unity3D platform. However, the platform does not support Augmented Reality by default, therefore, EasyAR was selected for this work. It is an Augmented Reality framework that can be used also in systems such as Windows, MacOS, Android and iOS, with the possibility of having its functionality used by both Unity3D and separate libraries.

III. TESTS WITH KIDS

To validate if the application of this project instigates behaviors related to Computational Thinking, three methods were used during the experiment. They were: (i) collecting data of application use on the mobile device, (ii) observations through videos of the activity and (iii) a questionnaire on the system usability. The data collection of the application and the observations by video were only

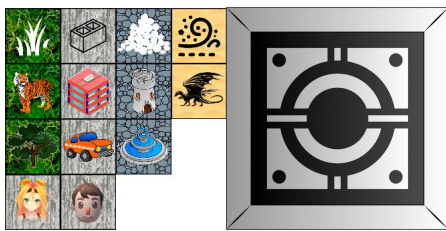


Figure 1. Tiles created for the application



Figure 2. Application graphical interface

collected during the execution of the activity, while the usability questionnaire was done at the end through questions to the students.

The application tests were done with three male students of different ages, being 9, 11 and 12 years old. They were from a local class, that was available to run the tests. This work application experiment was divided into four stages: (i) Presentation, (ii) Exploration, (iii) Activities and (iv) Questionnaire. They lasted about 50 minutes going through all the steps. The presentation stage lasted about 5 minutes, the exploration phase lasted about 30 minutes, and the activity phase lasted for about 10 minutes, finally the final 5 minutes were left to perform the questionnaire.

A. State Machines

Based on the methodology created by Santana [9], where the student's actions can be translated to state machines. Inside the application was left a state machine that collected student actions during activities. This machine can be divided into two, one for each screen, making it easier to analyze the results. The modeling of the machines can be seen in Fig. 3.

The states Programming, Debugging, Executing and Creating, are related to Computational Thinking, because they are problem-solving states. Lost is about losing the tracking and rest is when the child is not using the app. The state machine transitions in Fig. 3 are related to the following events during application usage:

T0: The child did not take any action for 10 seconds.

T1: The child lost the target tracking. It can be either all the tiles in the tracking screen, or the base tile in the rendering screen.

T2: The child tracked a tile on the tracking screen.

T3: The child added a command to the character on the rendering screen.

T4: The child removed a command from the character on the rendering screen.

T5: The child executed the command sequence given to the character.

Thus, by analyzing the state machine that was inside the application during the tests performed, it was possible to realize that during the tracking there were many traces of tracking loss, as can be seen in Fig. 4. This same machine also showed how the students stayed during the programming. There were many transitions occurred between the states of Programming, Executing and Refactoring, indications of Computational Thinking.

In Fig. 4, the number of entries made in each state is observable. It is possible to realize how much creation and lost had more impact than the moments related to programming.

The Fig. 5 graph was created by calculating the time difference of each transition made on the machine and adding each of these differences according to their state. It is realized that almost half the time of the tests were spent on map creation and almost half of this was lost in tracking losses, a higher time compared to the low spent on programming activities.

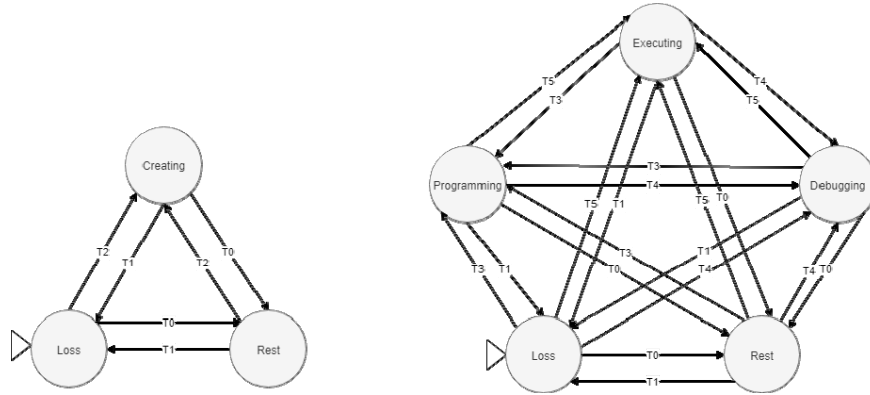


Figure 3. State Machines inside the application

Fig. 5, different from Fig. 4, also shows a high spent time in the rest state. Analyzing this with the footage, it is guaranteed to say that part of that time was spent with the device stopped because either the children were creating their maps, or it was the moment of the questionnaire, where nobody tampered with the application. Details of the footage can be seen in the next section.

B. Filming

For the filming details, the letters A B and C will be used to designate children of 9, 11 and 12 years respectively.

1) Presentation

The presentation was quick, about two minutes, since an explanation had already been made prior to the experiment. Child A was already very excited about the application and how it shows the map that it builds, already creating, at the time of the tutorial, the story that wanted to do with it, like wanting the character to move from the city to the forest with the Dragon. But the moment she was shown how to program, while she was impressed, she also wanted the character to "kill the dragon", but as there was only collision with the dragon, so she was a bit disappointed.

2) Exploration

At the beginning of the exploration, the child A started to

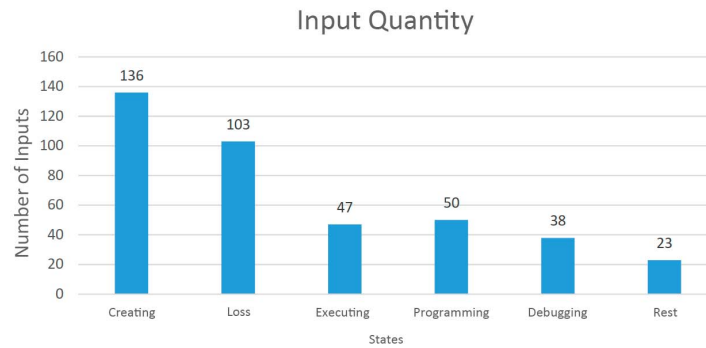


Figure 5. Input quantity graph

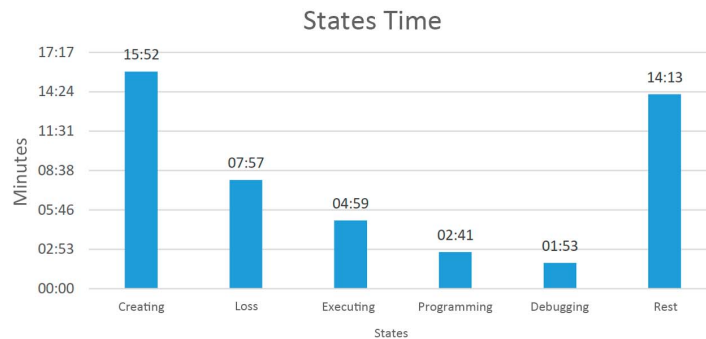


Figure 4. State stay time graph

build the map that he wanted to do while the other two continued to tinker with the application using the map in the tutorial, but soon began to assemble their own maps. Some doubts arose about what some tiles would be, like the tile of rocks, but not many.

Child A, set up a 3x6 map using all available dragons, using up pieces of other children. When he began to track him, he noticed the difficulties of the process, how to move very fast caused him to lose all the tracked tiles, so he gave up and set up a 2x3 map full of dragons. With a map of this size it was easy to track because the application was able to track all tiles at once. As soon as you went to the rendering screen and you can see the 3D map, child A really liked the result, but soon realized that it could not move the character on the map, since it was full of dragons as they cause a simple collision preventing the character from moving to the tile. She then asked if it was possible to walk on the grass, confirming this, she decided to reassemble her map with the grass tiles as well. Then child B, finished her map and wanted her one to be tracked. The map she created for it was 7x3, as soon as the upload to the tracking screen was finished, she tried to start tracking her own map.

She failed 6 times, and the last time she even tracked 20 of the 21 tiles needed. Because of this difficulty the child C wanted to try to track down his smaller map. It was a 3x3 map with a house, a fountain, a tiger and a tree as objects. She was able to quickly trace her map and render it at the base.

So she began to program his character to turn and go to the house he had done, but after executing the commands for the feat, he could not get the character out of there, because he had forgotten which direction he was pointing and there was only one direction which allowed him to leave the house, so it was necessary to use the reset button of the character, to put it in the initial tile that was wanted and to reprogram it.

She spent about 3 minutes programming the character and moving it around the map until there was a problem where the character was diagonally to the map. After the problem was resolved, the child stayed for another 3 minutes by moving the application until it was passed to child A again.

Child A had made a new 2x3 map, consisting of one dragon and the rest grass, after loading the tracking screen, it was possible to trace the map easily because of its size. She spent about 2 minutes playing with her created map, there being no problem during the process. As soon as she got tired, she moved on to child B to try his map again and left the area to do another lesson project.

After waiting for the app to load, child B could try to track her map. She stayed for about 6 minutes trying to track the 7x3 map that she had done unsuccessfully, and at some point she lost all her tracking and all the work, so she decided to cut her map to 4x3, which was still for her difficult to do, but in the end managed to get her map rendered. She spent another 2 minutes programming her character, getting a little surprised that the character could cross the tower in the process.

The total exploration activity lasted about 25 minutes. At the end of the process the planned activities began.

3) Activities

The activities were with a map of pre-established size but incomplete tiles. Children should complete it, track it, and then program it to a specified tile. Child A did not participate in the activities because she had started to work on another project and she was not interested, probably because she thought that her turn to touch the application was over.

The first activity occurred with a 3x3 map with the lower tiles already placed. The map was finished with concrete grass tiles having a tower and a tree in the center and should move the character to the tile of the house. When they finished tracking the map with some difficulties, they performed the execution, which occurred as programmed by them to move to the house.

The second activity occurred with a map of 4x3, with the tree and car tiles at the upper ends, a tiger and the character at the lower ends and a tree at the center. In Fig. 6 it is possible to see the before and after the map made in this activity.

This time the path to the goal was with simple tiles, almost separating a forest in the middle with a sandy road. As for programming and execution, it occurred without problems. After the end of the activities, the questionnaire for the children was done according to the established methodology.

C. Questionnaire

Finally, the questionnaire was done for the three children. The asked questions can be seen in table 1. Measures of ease in the use of the application, comfort, ease of activities and fun, were taken using the first 4 questions of the questionnaire with a likert scale [10]. This scale used from Likert values varies its values between "Not at all" (1), "I do not think" (2), "More or less" (3), "Yes, a little" (4) and "Certainly" (5). The responses given by the children can be seen in table 2.

According to their answers, there were no bad points in the application, but some of them showed a concern on some points. One of them responded with "more or less" in the questions of fun and comfort. She was the one who had difficulty with her map of more than 7 tiles in height, which she could not track. The other child gave good points for these questions but gave "More or less" for his amusement, this child being the one who made a simple map with house, tiger, tree and fountain. The last one did not participate in the activity, as she had other projects to do, so it did not answer



Figure 6. Children doing the second activity

TABLE I. QUESTIONS DONE TO THE CHILDREN

Number	Question
1	Did you find easy to use the app?
2	Did you felt confortable using the app?
3	The activities were easy?
4	Did you have fun using the app?
5	If you could add anything to the app, what it would be?
6	Would you like to say anything else?

TABLE II. CHILDREN ANSWERS TO THE QUESTIONS

	Child A – 9 years old	Child B – 11 years old	Child C – 12 years old
Q1	Yes, a little bit	More or Less	Yes, a little bit
Q2	Certainly	Yes, a little bit	More or Less
Q3	Did not participated	Yes, a little bit	Certainly
Q4	Certainly	Certainly	More or Less
Q5	Interactions with objects possibly allowing story creation	Improve tiles designs	More interaction options with objects, more beautiful tiles and new pieces
Q6	Nothing to say	Nothing to say	Nothing to say

this question, but the other ones found a good application.

Analyzing these answers, the great difficulty is due to the ease of use of the application, for issues previously raised, as the tracking. However, due to the low number of children used in the test, the results presented here may not be a factual representation of reality.

IV. CONCLUSIONS

Programming toys bring concepts of Computational Thinking with it, however, activities that stimulate this like the use of educational mats can cost a lot of time and money to be developed with different scenarios. The Augmented Reality being a technology that is in the pocket of the Brazilians majority, being in conjunction with smartphones, can allow the costs of creating pedagogical mats to be reduced. It is also estimated that it is important to investigate whether this type of technology allows students to have experiences like real ones due to their intrinsic immersion.

However, during the development of the tool, it was possible to perceive the limitations that the current technology possesses and how it makes difficult its use with educational methods outside the current teaching standard. One of the major limiters of this is the need for devices with powerful hardware for tracking real objects or images, with computers doing in average, up to ten simultaneous traces, while smartphones from four to six.

As for the analysis, tests were done with 3 children aged 9-12 years, and through them it was possible to perceive the

difficulty with the application in the moments of map tracking. This problem makes it very difficult for children to use this application actively, mainly because of the augmented reality high hardware requirement. But even with such difficulties, the actions of Computational Thinking were perceived during the tests performed. Assembling the maps with the physical tiles encourages the child to plan what she wants to do and how she wants to make her character move within the map following her schedule. Interacting with the physical and digital world has also motivated children to want each of them to take their turn from being able to use the application to have their own map made in augmented reality. Therefore, the application has the potential to stimulate Computational Thinking skills, but the current hardware limits prevent this application from causing enough impact and interest in children, as these problems create difficulties in application usability.

Finally, this work could present the current limits of technology and what are the possibilities that augmented reality can offer in the face of Computational Thinking. It is believed that with the advancement of technology in the future mobile devices can have a greater ease in their tracking, to the point where the limits presented here are not relevant and enable the application to be more accessible and easier to use.

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